



Lab Assignment || BCA || SEM VI || CA321 : Basics of Mobile Applications

1. Create an Application that can Toast a message "Welcome to CMPICA".
2. Create an Application to Print the name in Toast entered in EditText.
3. Create an Application to Input two numbers and show the addition answer in textView, Toast and in Edit Text by button.
4. Create an Application to understand Activity, Intent, Create sample application with login module. (Check username and password) open new activity and Print "Welcome username" manually without intent.
5. Create calculator application with two EditText to enter numbers, 4 Buttons to perform Addition, Subtraction, Division, Multiplication and display answer into toast. (Check for blank value and 0 value input) (Use proper controls)
6. Create an application to handle the view without creating its object and avoid using "findViewById" use the viewBinding.
7. Write an android application that will take basic student information (Student ID, Student Name, Student Branch, Student Contact No., Student Birth date) in one screen from user and display that information to other screen on submit button click of first screen) (Use proper controls).
8. Create an Application to show the Activity Life Cycle and show the messages in Logcate.
9. Create an Android application that should have 3 screens: Splash Screen Login Screen, Home Screen. When user start application for first time, Login screen should appear. User will login with Username: "Test" and Password: "TestUser". After successful login, user will redirect to home screen. Where Welcome [Username] should be display.
10. Write an Android application to convert string to upper case, entered by user into edittext on click of the button.
11. Create an android UI application to accept two numbers and the display the sum of the number using Button in EditText.
12. Write an android application to invoke built in application like contact, browser, google map and dialer. Operations are: By clicking on Visit the Website button, it should open "www.google.com" in android built in browser application By Clicking on make call button, it should open the phone dialer with a predefined number.
By Clicking on Show Map button, it should plot a specified location coordinates in the built in google map application.
By clicking on Choose Contact button, it should allow the user to select a particular contact and show the details as a toast message.
13. Design simple GUI application with activity and intents e.g. calculator use function to perform operations of calculator.
14. Write an application in android to manage the session of the user using SharedPreferences for Login Activity use functions to show stored ID password when application is launched.
15. Create an application to create a Registration form of the student for Ignite event and use the appropriate views. Also print data of the student on next activity. Keep only 2 or 3 competitions.
16. Create a Dice roller application with 1 to 6 numbers with images.
17. Create an application to show the alert dialog for application.
18. Create an application to show the autocomplete textview for country names.
19. Create an application to have All the menus in the application.
20. Create an application with a Spinner using Array Adapter for values.
21. Create a student photo and name listview with Base Adapter.
22. Create an Alert dialog for exit alert and on back press of mobile alert should be shown.



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23. Create an application to show the option to create, save and show the file and the file name in internal and external storage both.
24. Create an application to show the Navigation drawer and show the inbox, sent and deleted activities like Gmail.
25. Create an Application to show the tabs on one Activity and open each new fragment on each tab using tab layout and viewPager2 like WhatsApp table for status, chat and calls.
26. Create an Application to include Recycler view to show photos or names with card view and move it vertically and horizontally.
27. Create an application to have a CRUD operations using SQLite

Extra Assignment

28. Create an application to use firebase for Login Activities like Signup and Sign In.
29. Create an application to use Firebase for saving the data of an application.



Indicative Practical List

Assignment – 1

Sr. No	Questions										
Q1	Create an ASP .NET Application using MVC Architecture following requirements.										
	Application must showcase the techniques of Fetching of data from the database using Connected or Disconnected Approach.										
	Application must showcase the techniques of inserting the data into the database using Connected or Disconnected Approach.										
	Application must have well defined views to provide interaction to its functionalities.										
	Application must also be having functionalities to allow user to delete and update records, through a view.										
	Note: For reference you may use below provided structure to create table in database or it is up to you to select one of your choice.										
<table><tr><th colspan="5">TBLFoods</th></tr><tr><th>ID</th><th>Name</th><th>Cuisine</th><th>Price</th><th>Quantity</th></tr></table>		TBLFoods					ID	Name	Cuisine	Price	Quantity
TBLFoods											
ID	Name	Cuisine	Price	Quantity							

Assignment – 2

Create an ASP .NET WEB API Application to fulfill following requirements.

Questions

1. Create a Web API to fetch all the data from the database and return it. (Use Attribute based routing; to allow consumption of the API, for the consumer)
2. Create a Web API to fetch all the data of specific cuisine, as per the request of consumer.
3. Create a Web API to get data from the consumer and insert it in the Database.
4. Add validation in model with Data Annotations and make Name and Quantity a Required Field, create a custom exception to generate a proper error message and avoid data insertion, if the same is not followed by the consumer while consuming the “Data Insert API” (3).
5. While inserting the data in “Data Insert API” (3), check if Quantity is not less than 0, for such case; create a custom exception to return proper message and avoid data insertion.

Note: Use the provided structure to create tables in database.

TBLFoods				
ID	Name	Cuisine	Price	Quantity

Note: Use Entity Framework for Database Operations.

Lab Assignment || MCA || SEM II || CA871 : Advanced Mobile Programming

Assignment-1

1.	Write a program to display "Welcome to Flutter" in the center of the screen. Set up an Android Virtual Device for your project and run your program on this virtual device.
2.	Write a program to display "Flutter works on Web!" in the center of the screen. Set up Google Chrome browser as the target device and run your program in the browser.
3	Write a program to display at least four built-in icons on the screen.
4.	Write a program to display at least four images-each of different type such as JPEG, PNG, GIF etc. Also display an image available on Internet using its URL.
5	Replace default app icons for a Flutter project manually.

Assignment-2

1.	Write a program which takes two numbers as input. Provide four buttons for addition, subtraction, multiplication and division. Display output of the arithmetic operation in another Text.
2.	Display a group of five checkboxes and display all the currently selected items in a Text as and whenever any checkbox selected or deselected.
3	Re-write program. 1 by replacing buttons with four radiobuttons for addition, subtraction, multiplication and division.
4.	Write a program to create two tabs with different background colors. First tab should display three images with the help of a Row widget and the second tab should display three images with the help of Column widget.
5	Write a program to create an app with four screens. Each screen must have three buttons to re-direct to one of the three other screens. Use named routes.

Assignment-3

1.	Write a program to display three horizontal scrollbars: one for each of Principle, Rate of Interest and Term (i.e. no. of years). Calculate and display simple interest depending on the current value of principle, rate of interest and term.
2.	Create a scrollable dynamic list of 25 checkboxes with the help of ListView and builder() method.
3	Create a dynamic grid of three columns. Display seven icons in this grid.
4.	Write a program to create a drawer with three options: Login, Greet and Close. Create three dart files: Login.dart, Greet.dart Tapping on Login should display screen for login from Login.dart Tapping on Greet should display screen with "Have a good day" message from Greet.dart. Tapping on Close should close the drawer.



Assignment-4

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|----|---|
| 1. | Write a program in which TextField widget is refactored. |
| 2. | Write a program to play a sound file using a Flutter app. |

Assignment-5

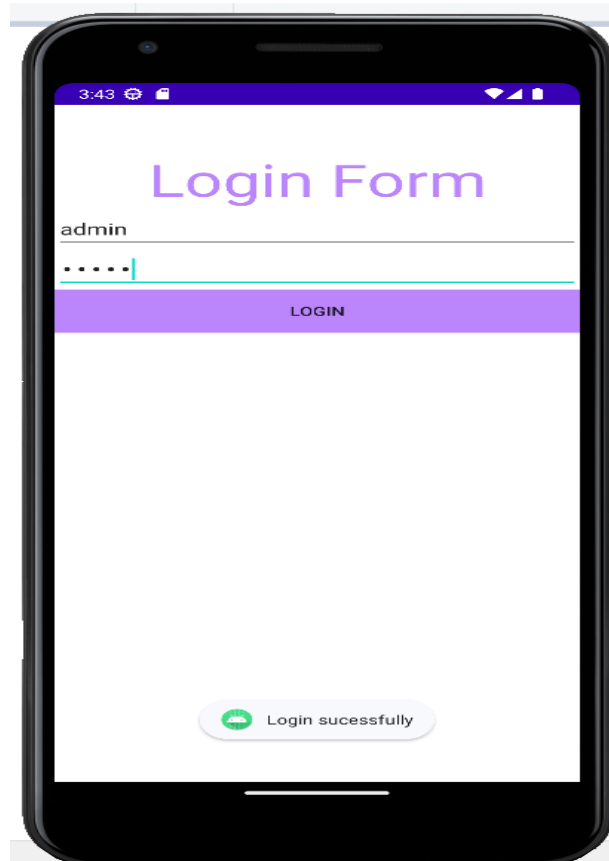
- | | |
|----|---|
| 1. | Write a program to read data from a JSON file and display the same using a Flutter app. |
| 2. | Write a program to write data to a JSON file using a Flutter app. |
| 3. | Write a program to display current location of the device using a Flutter app. |
| 4. | Write a program with two screens: The first screen will have an empty text and a button to move to second screen. The second screen will have a textbox and a button to move to the first screen. The value entered in the textbox should be displayed in the text on the first screen. |
| 5. | Write a program to perform create, retrieve, update and delete data from an online REST API. |

Assignment-6

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|----|--|
| 1. | Write a program to show animation using Hero widget in a Flutter app. |
| 2. | Write a program to create and display BottomSheet in a Flutter app. |
| 3. | Write a program to display real-time clock using StreamBuild in a Flutter app. |
| 4. | Write a program to perform create, retrieve, update and delete operations on data available in Firebase. |

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1. Create android UI application to accept two numbers and the display the sum of the number using Button and EditText.
2. Create an Android application to demonstrate the Activity LifeCycle.
3. Create sample application with login module. (Check username and password). If user name and password is “admin” display message “Login successfully” on the screen.



4. Create calculator application with two EditText to enter numbers, 4 Buttons to perform Addition, Subtraction, Division, Multiplication and display answer into toast. (Check for blank value and 0 value input) (Use proper controls)
5. Write an android application that will take basic student information (Student ID, Student Name, Student Branch, Student Contact No., Student Birth date) in one screen from user and display that information to other screen on submit button click of first screen) (Use proper controls)
6. Create an Android application that should have 2 screens: Login Screen, Home Screen. When user start application for first time, Login screen should appear. User will login with Username: “admin” and Password: “admin”. After successful login, user will redirect to home screen. Where Welcome [Username] should be display. (Use explicit Intent)
7. Open CHARUSAT website using Implicit Intent.

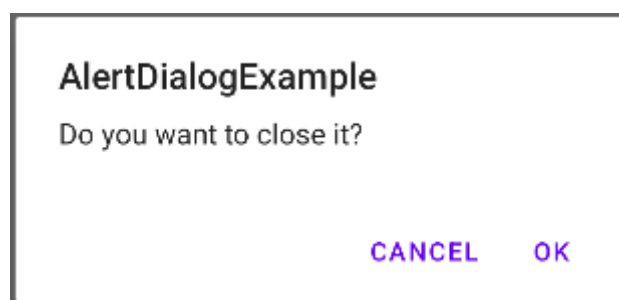
Lab Assignment || B.sc(IT) || SEM VI || CA610 Mobile Application Development

8. Create Application of Alert Dialog Box contains following three components.

Title

Message

Buttons (that trigger Negative and Positive Actions)



9. Display Toast message by clicking on Image Button and open CHARUSAT website using webview.

10. Write a program using Radiobutton and display message in Toast.



11. Write a program using Checkbox and display message in Toast



12. Write a program for Toggle Button and switch and print the status of the button in Toast.

13. Display the states name in AutoCompleteTextView and MultiAutoCompleteTextView.

14. Display your Birthdate in TextView using CalenderView, Date Picker, Time Piker.

15. Display Charusat website using WebView

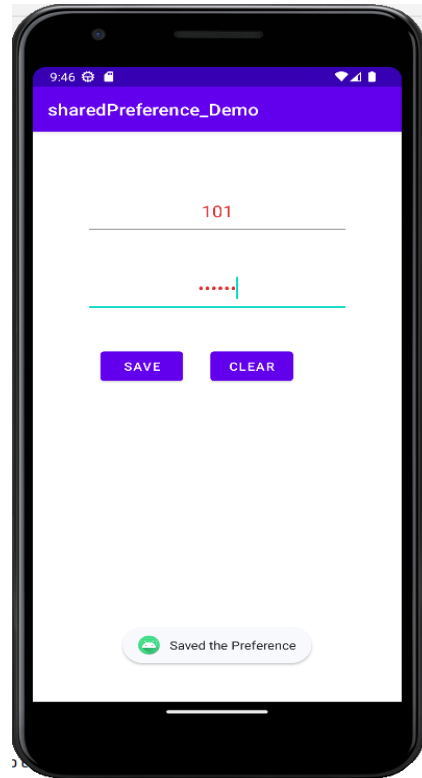
16. Display any Video using VideoView

17. Create one Progressbar

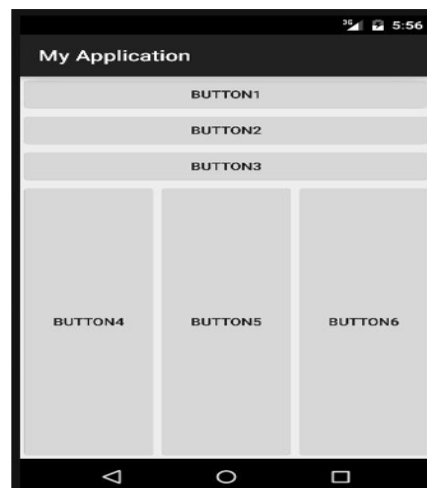
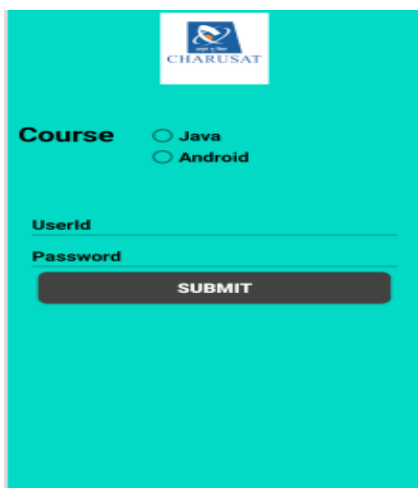
18. Generate feedback using Ratingbar.

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19. Write an application in android to save User ID and password in login form using SharedPreferences.



20. Design the Simple GUI using proper Layout also set the background color using Drawable Resource File.

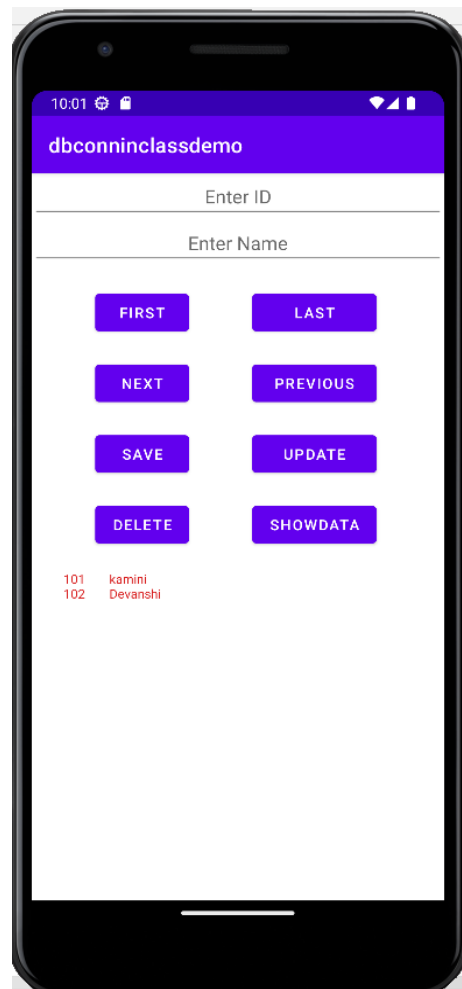


21. Create following GUI using Table Layout and add the product in the SQLite Database

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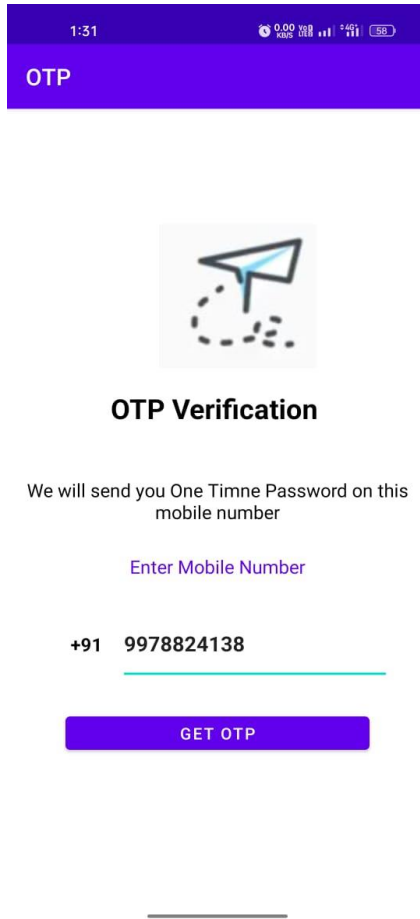
22. Write an application of Crud operations using SQLite database.



23. Create an application to use firebase for Login Activities like Signup and Sign In.

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24. Create an application to use Firebase for saving the data of an application.
25. Create an Application to generate OTP on the given no using login form.



1:31 0.00 KB/s 4G+ 58%

OTP



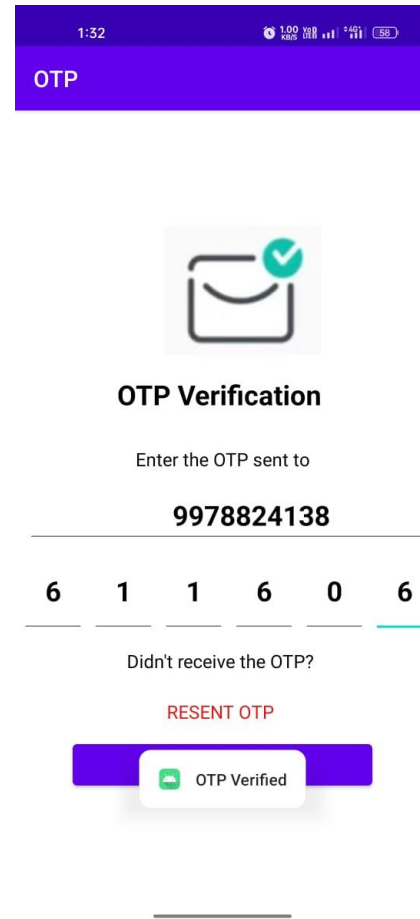
OTP Verification

We will send you One Time Password on this mobile number

Enter Mobile Number

+91 9978824138

GET OTP



1:32 1.00 KB/s 4G+ 58%

OTP



OTP Verification

Enter the OTP sent to

9978824138

6 1 1 6 0 6

Didn't receive the OTP?

RESENT OTP

OTP Verified